14/3 vs 14/4 AC90 — nursery rack lighting

One cable feeds 2 lit racks (each ~8,8 A on its own 15 A circuit). 10 lit racks → 5 pairs, ~30 m per pair. Prices before tax, Home Depot / Gescan, Sept 2026.

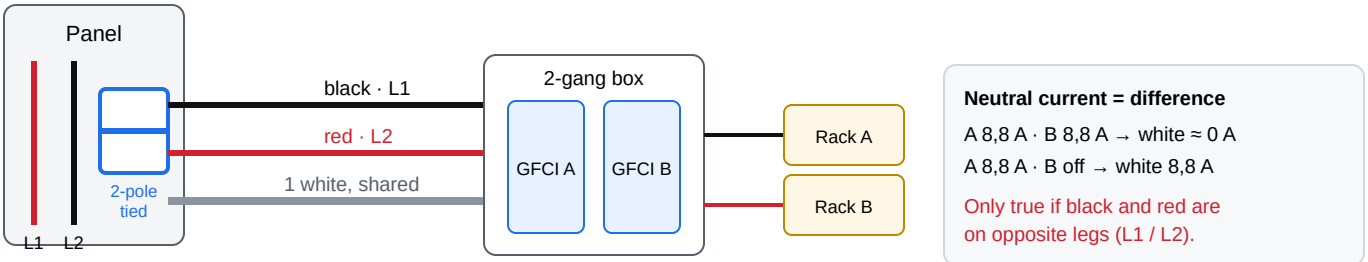
Verdict

	14/3 shared neutral	14/4 separate neutrals
Conductors	black + red + **1 white** + bare	black + red + **2 white** + bare
Cable, 150 m	**~495 $ (Home Depot, 3,30 $/m)**	~1 145 $ (Gescan, 7,64 $/m)
vs 14/2, 1 run per rack (~950 $)	**−455 $**	+195 $
Breaker per pair	2-pole, handle-tied (required)	**tandem 15/15 (as today)**
Panel slots, 5 pairs	10	**5**
Wiring mistakes that matter	both hots on the same leg · GFCI neutral through LOAD	**swapped neutrals between circuits (GFCI trips)**
Loose neutral	both racks in series on 240 V → can burn timers + LED drivers	**only that rack goes dark**
Availability	**Home Depot, in stock**	wholesaler only

Pick 14/3 if the panel has 10 free slots — cheapest, 5 pulls. **Pick 14/4** only to avoid the shared neutral, at ~650 $ more than 14/3. **Stay on 14/2** if the panel is tight.

How each one is wired

14/3 — multi-wire branch circuit



14/4 — two independent circuits

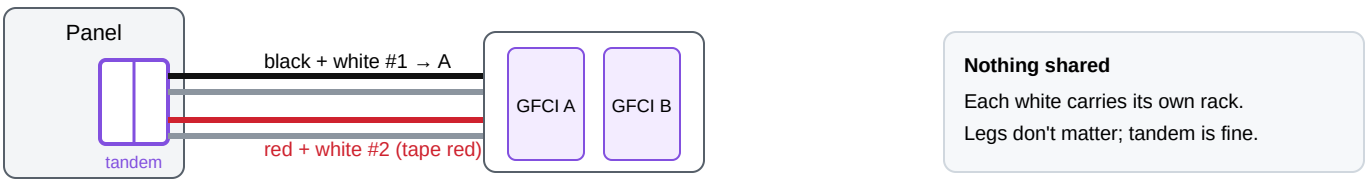


Figure 1 — Same box, same 2 GFCIs. The difference is one shared white (14/3) vs two separate whites (14/4).

What can go wrong with 14/3

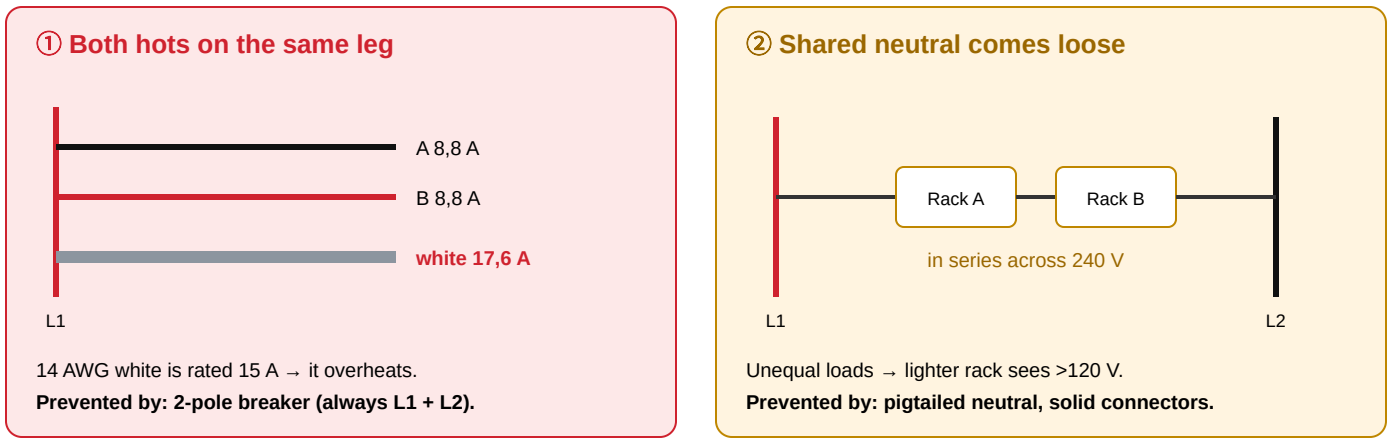


Figure 2 — Both are installation errors, not design flaws. A careful electrician avoids both.

Timers in the 2-gang box

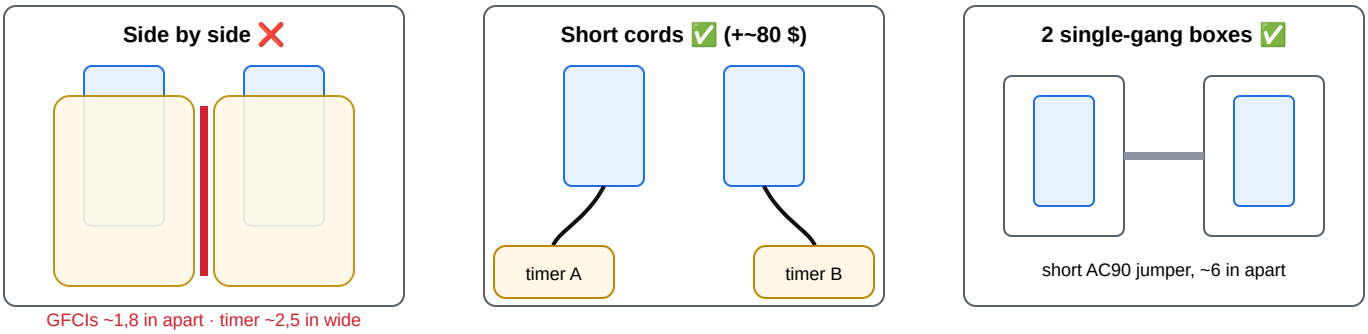


Figure 3 — Two plug-in timers don't fit side by side. Cords or two boxes solve it with either cable.

Cost for 10 lit racks (5 pairs)

Setup	Cable	Cords	Boxes + plates	vs 14/2 today
14/2, 1 box per rack (today)	300 m · ~950 $	—	~50 $	—
14/3 2-gang per pair	**150 m · ~495 $**	**~80 $**	**~50 $**	**≈ −375 $**
14/4 2-gang per pair	150 m · ~1 145 $	~80 $	~50 $	≈ +275 $

Questions for the electrician

- How many free slots in the panel? 14/3 needs 10, 14/4 and 14/2 need 5.
- Is the service 120/240 V single-phase? On 208 V three-phase, a 14/3 neutral carries full load plus LED-driver harmonics.
- Does the inspector accept AC90 in the nursery (dry-location cable)? Rodents rule out NMD90 / NMWU without conduit.
- 14/3: 2-pole handle-tied breaker, neutral pigtailed to each GFCI LINE, never through LOAD.